Zhexiang (Bruce) Li

bruceli@ucla.edu +1 (831) 239-2728 Los Angeles, CA Website GitHub LinkedIn

RESEARCH INTERESTS

Multimodal learning for health; efficient attention and visual token routing; interpretable molecular and spectrum modeling.

EDUCATION

University of California, Los Angeles Sep 2026 - Jun 2028 (expected)

M.S. in Computer Science

University of Southern California May 2026

B.S. in Computer Science · GPA 3.76 / 4.00

RESEARCH EXPERIENCE

Graduate Researcher | Health Intelligence Lab (HAIL), UCLA Aug 2026 - Present

Advisor: Prof. Yuzhe Yang

- Investigate multimodal foundation models for diabetes, integrating continuous glucose monitoring with insulin, meals, activity, and patient context.
- Built patient-disjoint MetaboNet benchmarking and audit pipelines. Evaluated GluFormer, T1DFormer, and custom predictive-state and Transformer architectures at 30/60/120-minute forecasting horizons.

USC Drug Discovery Research Spring 2026

Research with a Ph.D. student in Prof. Yan Liu's group

- Studied SARS-CoV-2 antiviral discovery through reproducible benchmarks of molecular descriptor-based models, Chemprop, and hybrid approaches across seven Therapeutics Data Commons datasets, using scaffold splits.
- Analyzed descriptor importance and SHAP attributions to examine molecular features associated with model predictions.

PUBLICATIONS & MANUSCRIPTS

MedTokenBudget: Lesion-Preserving Token Routing for Dermoscopic Image Classification 2026

Zhexiang Li (sole author). Preprint · arXiv:2609.30613

Developed lesion-supervised post-backbone token routing. At a 30% token budget on ISIC 2019, achieved 80.38 ± 0.18% accuracy versus 76.93 ± 0.03% for ToMe across three seeds; the frozen backbone remains unchanged.

Mass-Conservation Token Transformers for Auditable MS/MS Spectrum Retrieval 2026

Manuscript submitted to AAAI 2027

Represented mass-conserving fragment pairs as sparse evidence tokens for spectrum retrieval. Used closest-mass and overlap-hard negatives, alongside evidence-removal tests, to examine whether the exported evidence affects ranking.

Efficient Attention via Pre-Scoring: Prioritizing Informative Keys in Transformers 2025

Zhexiang Li, Haoyu Wang, Yutong Bao, David Woodruff. Under review · arXiv:2505.11040

First-author work advised by Prof. David P. Woodruff (CMU). Proposed pre-scoring and implemented informative-key selection in Triton. Integrated with HyperAttention, reduced ChatGLM perplexity from 12.0 to 9.5 at 131k-token contexts under a fixed interaction budget.

Game Theory in Companies 2023

S. Li, Z. Li, Z. Liu. Advances in Economics, Management and Political Sciences, 16(1), 357-366. DOI

Advised by Prof. Ming Gu (UC Berkeley). Led a team modeling game-release decisions as two-player normal-form games and enumerating Nash equilibria with Python and Gambit.

PROFESSIONAL EXPERIENCE

AI Engineer Intern | PMAOS / Next Big Seek Jul - Aug 2026

- Designed short-video recommendation using bidirectional user-video interest propagation, exposure budgets across multiple timescales, and fatigue controls to manage repeated recommendations.
- Developed 11-language audio-visual affect inference with evidence fusion and a fixed-point decision layer that can abstain or report conflicting modalities.
- Screened sub-10 mm piezoelectric robot architectures and specified a PZT-steel cantilever experiment for a 9 mm candidate; work concluded at experiment and procurement planning.

Full-Stack Engineer | Terlingua Threads, Remote Jan - May 2026

- Built Shopify-integrated pages and Google Maps features; improved responsive layouts, accessibility, SEO, and maintainable content structures.

Co-Founder & CTO | CrAIg Capital, Los Angeles, CA 2025 - Mar 2026

- Led product and engineering for AI-assisted housing debt collection, including data pipelines and automated workflows.
- Set technical direction, hiring priorities, and delivery standards during startup development.

AI Engineer Intern | Piezolab Medical, Lake Forest, CA Jun - Aug 2025

- Built an LLM-assisted pipeline to ingest, clean, and cluster 5,000+ medical-malpractice documents for quality-of-care review.
- Developed agent workflows, experiment checklists, and acceptance tests for vascular-care research.

Business Development Intern | Hangzhou Haoan Jianke Technology, Hangzhou Jun - Aug 2024

Gathered requirements for a NetEase CodeWave internal platform, helped recruit engineers, and contributed to program structure and interface planning.

Application Engineer Intern | Synaptics, San Jose, CA Jun - Aug 2023

Validated product firmware and built an ESP32-to-Alibaba Cloud pipeline that published I²C sensor data as MQTT topics for internal dashboards.

Investment Analyst Intern | GUDU Medical, Remote Jan 2021 - Present

Supported MedTech deal screening, market analysis, due diligence, financial modeling, and valuation.

SELECTED PROJECTS

Fridge Manager 2026

Next.js · Supabase · howtoeat.food

Built meal and grocery planning around available ingredients, leftovers, recipes, dietary preferences, and cooking time.

AI Doctor WeChat 2026

FastAPI · Knowledge retrieval · WeChat Mini Program

Built a clinical teaching and collaboration application connecting WeChat and web workspaces to a FastAPI backend with knowledge retrieval and patient-intake workflows.

AI Business Card Hub

Jan 2026 - Present

iOS · Vision parsing · OCR · Source code

Built photo-to-contact extraction with OCR fallback, multi-photo review, bilingual records, and searchable people and company profiles.

TEACHING

Tutor | Introduction to Machine Learning, UCSC

Spring 2023

Prof. Lise Getoor

- Held office hours and discussion sessions on logistic regression, SVMs, and tree models.
- Co-designed Jupyter Notebook assignments and contributed to the coding autograder.

Assistant Coach | Shanghai Bosidi Education

2014 - 2018

VEX Robotics & Olympiad in Informatics

Supported robot design and algorithmic problem-solving instruction, scrimmages, and events.

HONORS & ACTIVITIES

Viterbi Dean's List, University of Southern California: Fall 2024 - Spring 2026.

TIE Hub: Co-Founder Speed Date volunteer (Sep 19, 2025); five-day startup hackathon participant, including prototype development, venture design, and pitching (Sep 2025).

Noah Business School: Entrepreneurship, financing, and Silicon Valley company visits (2025). **USC EVMA:** Member.

TECHNICAL SKILLS

Programming: Python, C/C++, JavaScript/TypeScript, SQL

Machine learning: PyTorch, Triton, RDKit, Chemprop; multimodal modeling, efficient attention, recommendation, and molecular prediction

Engineering: FastAPI, Next.js, Supabase, Linux; automated testing, static type checking, and data pipelines

Updated October 2026 · Full experience archive & application resumes